Machine-checkable lower-bound certificates for the circuit-level distances of bivariate-bicycle syndrome-extraction circuits

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The circuit-level distance of a syndrome-extraction circuit is the smallest number of fault mechanisms that flip a logical observable without firing a detector. We give machine-checkable certificates for the circuit distances of four bivariate-bicycle circuits at the scale of the [[144,12,12]] gross code, written by a branch-and-bound search on the same-type projection of the detector error model and re-derived by a separately written checker whose proof calculus is proven sound in Lean 4. IBM's depth-8 cycle has distance 10 for every number of rounds from 2 up, as G. Pan's public research record also reports (certified to twelve rounds, argued beyond); our lower bound for every number of rounds rests on certificates, a hand proof and script-checked hypotheses. The three-colour cycle of Strikis, Browne and Beverland has 11, the value G. Pan derived from their reduction and exhaustive search. Cheng, Guo and F. Pan certify both with an exact solver, the second also for every number of rounds. For Shaw and Terhal's two morphing circuits, G. Pan's record already reports 6 and 12 for every number of rounds, by exhaustive search without certificates; we add certificates and a fold theorem whose hypotheses are machine-checked. New here are exact values for 14 morphing circuits we constructed with the code parameters of Shaw and Terhal's 2026 follow-up. Our projected mechanism sets equal those of G. Pan's own circuits under his noise model at 1 to 3 rounds. Every certificate the results rest on was re-checked by the committed checker. For 4 small models, none of them a gross-code model, the Lean kernel checks the exported model's distance. Every-round statements rest on two hand proofs and readings of the builders' code, reviewed only by AI agents and not formalized; their finite hypotheses are machine-checked. Distances are properties of the stated fault placement, not error rates.

I. INTRODUCTION

A syndrome-extraction circuit can have a smaller distance than the code it measures. A fault on an ancilla can spread through the CNOTs onto several data qubits, and faults in different rounds can combine. The quantity that bounds a memory experiment is the circuit-level distance d_{circ}: the smallest number of fault mechanisms whose combined effect fires no detector and flips a logical observable. It limits every decoder. Split a minimum-weight undetectable logical fault set into two halves; they have the same syndrome and differ by a logical operator, so any decoder fails on one of them, and the logical error rate is at least of order $p^{\lceil d_{\mathrm{circ}}/2 \rceil}$ for independent faults of probability p.

For bivariate-bicycle (BB) codes [1], several extraction circuits have been proposed: IBM's depth-8 cycle for the [[144,12,12]] gross code [1]; the three-colour cycle of Strikis, Browne and Beverland [2]; and the ancilla-free morphing circuits of Shaw and Terhal [3], with a 2026 follow-up [4]. In the papers that introduced them, the circuit distances are upper bounds found by heuristic search, or are not reported: $d_{\mathrm{circ}} \leq 10$ for IBM's cycle, from a BP-OSD search [1, 5, 6]; 11, conjectured from a heuristic search, for the three-colour cycle; ≤ 6 and ≤ 12 for the two morphing circuits, from BP-OSD searches; and, for the 2026 circuits, end-cycle code parameters only, with the statement that the circuit-level distance was not checked (their App. C.4). An upper bound needs one fault set. A lower bound needs an argument about every lighter fault set.

For smaller BB circuits exact solvers have supplied that argument: Aghababaie Beni, Higgott and Shutty [7] verified with Stim's SAT export [8] and a MaxSAT Evaluation 2024 solver that IBM-style circuits for the [[72,12,6]] and [[90,8,10]] codes have circuit distances 6 and 8 (their App. D), and Webster, Jacob and Higgott [9] report that integer programming finished within 8 hours only on the first circuit of a BB circuit family (their Sec. 5.5.3). For the gross-code-scale circuits, other works appeared while ours was in progress. Crocker [10] showed by an exhaustive branch-and-bound search, built on the same three reductions we use (same-type projection, anchoring by translations, anchoring by time shifts), that IBM's cycle has $d_{\mathrm{circ}} = 10$ in both bases for twelve noisy rounds followed by two noiseless ones, without a machine-checkable certificate. Under SD6 noise, Guancheng Pan [11] proved that the three-colour cycle has $d_{\mathrm{circ}} = 11$ for every number of rounds, from the reduction of Strikis, Browne and Beverland (their Prop. 3 and Cor. 1) and an extended-code distance of 11 found by two independent exhaustive searches, of which a bound of 9 is certified; and constructed interleaved gross-code circuits with $d_{\mathrm{circ}} = 12$, whose infeasibility claims are backed by proofs checked with DRAT and VeriPB tools. For IBM's cycle his released manuscript reports $d_{\mathrm{circ}} = 10$ for up to three rounds by exact enumeration, and his research record goes further on his repository's main branch after that release: record E-037 [12]

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gives VeriPB certificates that no undetectable logical set of weight ≤ 9 exists at every $R = 1, \ldots, 12$ in both bases, under SD6 and under a model rebuilt from IBM's `decoder_setup.py` (the certified formula is strengthened by minimality and symmetry-breaking constraints whose soundness rests on a written argument), and record E-043 [13] extends $d_{\text{circ}} = 10$ to every number of rounds under SD6, from his bulk criterion checked by two solvers, a direct search up to three rounds, and a persistence argument for the witnesses. Record E-045 [14] does the same for his re-implementation of IBM's `decoder_setup.py` model, R noisy cycles followed by two noiseless ones: $d_{\text{circ}} = 10$ for every $R \geq 1$ in both bases, by a written time-translation, periodicity and compression argument that reduces every R to the range E-037 certifies, with its finite premises checked by exact frame simulation; the same record certifies, with VeriPB, distance 10 for the twelve-round SI1000 circuit distributed with Tesseract. Of these steps, the VeriPB runs of E-037 and E-045 produce proof logs; the solver runs of E-043 and the every-R steps of E-043 and E-045 do not. Pan's released research record, public with his first release, also settles the two morphing circuits we study: the [[144,12,12]] circuit has $d_{\text{circ}} = 12$ for every number of rounds, in both bases and with noisy preparation and readout (record E-017 [15], Secs. 3, 4 and 8), and the [[72,12,6]] circuits have $d_{\text{circ}} = 6$ (record E-019 [16], Sec. 2.1, exhaustive on four-round circuits, with every number of rounds from per-circuit fold checks, Sec. 5). Both records rest on exhaustive searches by independently written programs and on a computational check of a fold premise, without certificates. Bin Cheng, Naixu Guo and Feng Pan [17] (below, Pan alone is Guancheng Pan) compute exact code and circuit distances at scale. In their sense a distance is certified when a lower bound from completed thresholds of a solver whose completeness they prove meets an explicit logical error replayed in Stim; their records give each threshold's search log and the witness, not a proof object that a third party can check without repeating the search. For IBM's cycle they certify $d_{\text{circ}} = 10$ in both bases for one noisy cycle of the memory instruction of IBM's architecture study, between an error-free code state and a noise-free final round, and for twelve noisy cycles of the circuits of IBM's public code; their time-collapsed relaxation, a lower bound for every number of rounds, gives 8 in memory-X and 9 in memory-Z, and the lifting that raises it to 10 is for a fixed number of rounds (their Methods and Supplementary Sec. 9.2). For the three-colour cycle they confirm the conjectured 11 in both bases (their Supplementary Table 8), and the records in their repository give their time-collapsed bound, a lower bound for any number of rounds (their Supplementary Sec. 5.4), as 11 in both bases too. Their paper does not treat morphing circuits. Wills *et al.* [18] describe "a new idling circuit for the gross code", constructed with Shaw–Terhal morphing, that "is believed to have circuit distance 12", and simulate its logical error rate; the circuit is not specified there, so we have not compared it with ours. If it is the morphing circuit of Ref. [3], its end-cycle code has the gross code's parameters but is not the gross code (Sec. VI). We make no priority claim for any of these values, and none about the order of discovery.

This paper contributes certificates, a cross-check of our circuits against Pan's own, and new values (Table I):

1. *Certificates checked under a verified calculus.* A branch-and-bound search on the same-type projection of the detector error model, anchored on translation orbits with an affine map of the observables, writes a certificate of about one bit per node, and a separately written checker re-derives everything else from the model (Sec. III). The checker's calculus is proven sound in Lean 4 [19] with Mathlib [20]; the formalization found inputs on which the C checker accepted a false claim; and for 4 small models the Lean kernel checks the distance of the exported model itself (Sec. V).

2. *Certificates for values already reported.* IBM's cycle has $d_{\text{circ}} = 10$ for every $T \geq 2$ in both bases, 2 below the code distance. Pan's records certify the lower bound up to twelve rounds and extend it to every number of rounds, under SD6 (E-043) and for his re-implementation of IBM's fault model (E-045), and Cheng, Guo and Pan certify the value at fixed numbers of rounds with their exact solver; ours rests, for every T, on certificates re-derived by a separately written checker and on Theorem 1, whose finite hypotheses are checked by script. The three-colour cycle has $d_{\text{circ}} = 11$ for every $T \geq 2$; Cheng, Guo and Pan certify the value with their exact solver, and their time-collapsed bound, which they prove holds for every number of rounds, is already 11 in both bases; ours is, among the works we found, the first lower bound of 11 for every number of rounds that rests on certificates a separate checker re-derives and on Theorem 1. The two morphing circuits keep their end-cycle code's distance, 6 and 12, for every $T \geq 1$, the values of Pan's record; ours are the first certificates for them that we found. Our projected mechanism sets equal those of Pan's circuits (Sec. II C), so these agreements are on one object.

3. *New values.* For 14 circuits we constructed with the code parameters of the 2026 follow-up we give exact values, 4 of them kernel-checked at one round (Table II). On them the CNOT order alone moves the distance by up to 2. That the CNOT order changes a morphing circuit's distance is not new: Pan's records show it on the codes of Ref. [3] [16].

4. *Every number of rounds.* Two theorems reduce every T to finitely many models: a window formula for the IBM and three-colour cycles, and a fold theorem for the morphing circuits (Sec. IV).

TABLE I. Exact circuit-level distances, memory-Z / memory-X, for the fault placement of Sec. II C. *Prior value*: the value reported in the circuit's source, then the value of Pan's research record or manuscript, for IBM's cycle certified up to twelve rounds and argued for every number of rounds [12–14], for the three-colour cycle for every number of rounds [11], for the morphing circuits by exhaustive search [15, 16]; for the last row, the exact value of Ref. [7]; each in its source's own noise model. Ref. [10] gives 10 for IBM's cycle at twelve noisy rounds by exhaustive search, and Ref. [17] certifies 10 for IBM's cycle and 11 for the three-colour cycle at fixed numbers of rounds with an exact solver. *Lower bound certified*: the rounds T at which a certificate excludes every lighter solution. For IBM's cycle, the three-colour cycle and the [[72,12,6]] morphing circuit these are time-shift certificates (Sec. IV A): each excludes every *irreducible* lighter solution that starts in round 0 or 1, and the every-round value follows from them only with rule S and Theorem 1 (or Theorem 2). *Witness verified*: the rounds at which a fault set of weight d_{circ} was verified on the full detector error model. *Holds for*: the range covered by the theorems of Sec. IV; every entry there is proven from machine-checked premises. The last row is a smaller instance of IBM's schedule, memory-Z only; its certificate is regenerated and checked by the test suite and is not stored.

circuit	code (mid-cycle)	prior value	d_{circ} (Z / X)	lower bound certified	witness verified	holds for
IBM depth-8 [1]	$[[144, 12, 12]]$	≤ 10; 10 [12–14]	10 / 10	$T = 1, \ldots, 11$	$T = 2, \ldots, 6$	$T \geq 2$
three-colour	$[[144, 12, 12]]$	11 (conj.); 11 [11]	11 / 11	$T = 1, \ldots, 12$	$T = 2, \ldots, 12$	$T \geq 2$
morphing	$[[72, 12, 6]]$ ($[[144, 12, 12]]$)	≤ 6; 6 [16]	6 / 6	$T = 1, \ldots, 7$	$T = 2, \ldots, 7$	$T \geq 1$
morphing	$[[144, 12, 12]]$ ($[[288, 12, 18]]$)	≤ 12; 12 [15]	12 / 12	$T = 1, 2$ (fold: $T \leq 13$)	$T = 1, \ldots, 13$	$T \geq 1$
IBM-style	$[[72, 12, 6]]$	6 (exact) [7]	6 / –	$T = 2$	$T = 2$	–

TABLE II. Circuit-level distances of morphing circuits we constructed with the code parameters reported in Ref. [4] (Sec. VI D). *Degree*: the largest number of distinct CNOT partners of any qubit over the cycle. For the standard schedules, A and B list the polynomial terms in schedule order, so circuits that differ only in that order are different CNOT orders on the same code. The tree circuits are homomorphism-based circuits (their Def. 10), each fixed by one contraction tree (their Sec. 2.4); they share A, B (listed in index order) and the code and differ only in that tree, whose layers list parent→child pairs on the support vertices $0, \ldots, 5$ (A's terms $0, \ldots, 2$, then B's): `t30a`: $1{\to}4, 5{\to}2 \mid 0{\to}5, 3{\to}1 \mid 0{\to}3$; `t30a2`: $1{\to}4, 5{\to}2 \mid 0{\to}5, 3{\to}1 \mid 3{\to}0$; `t30b`: $5{\to}0, 1{\to}4 \mid 3{\to}1, 2{\to}5 \mid 2{\to}3$; `t30b2`: $5{\to}0, 1{\to}4 \mid 3{\to}1, 2{\to}5 \mid 3{\to}2$. *Loses*: end-cycle distance minus d_{circ}. *Rounds*: *every* T is proven from machine-checked premises (Theorem 2; that their builder emits the theorem's circuit for every T is read from its code and checked at every $T \leq 32$, Sec. IV B); MC is machine-checked for the rounds shown only. *Kernel ($T = 1$)*: the distance of the one-round projected model is also a theorem checked by the Lean kernel; the witness's lift to the full model is checked outside Lean (Sec. V). For a tree circuit the kernel result covers $T = 1$ alone, since Theorem 2 does not apply to it.

circuit	end-cycle code	degree	schedule	A	B	d_{circ} (Z / X)	loses	rounds	kernel ($T = 1$)
`c42a`	$[[42, 6, 6]]$	4	standard	$(x, 1, x^{10})$	$(1, x^7)$	6 / 6	0	every T	Z, X
`c42a2`	$[[42, 6, 6]]$	4	standard	$(x, x^{10}, 1)$	$(1, x^7)$	4 / 4	2	every T	–
`c42b`	$[[42, 6, 6]]$	4	standard	$(1, x^3, x^9)$	$(1, x^7)$	5 / 5	1	every T	–
`c42b2`	$[[42, 6, 6]]$	4	standard	$(1, x^9, x^3)$	$(1, x^7)$	4 / 4	2	every T	–
`c48a`	$[[48, 4, 8]]$	4	standard	$(1, x, x^5)$	$(1, x^{15})$	7 / 7	1	every T	–
`c48a2`	$[[48, 4, 8]]$	4	standard	$(1, x^5, x)$	$(1, x^{15})$	6 / 6	2	every T	–
`c48b`	$[[48, 4, 8]]$	4	standard	$(1, x, x^5)$	$(1, x^{21})$	6 / 6	2	every T	–
`c48b2`	$[[48, 4, 8]]$	4	standard	$(1, x^5, x)$	$(1, x^{21})$	6 / 6	2	every T	–
`c48c`	$[[48, 4, 8]]$	4	standard	$(1, x, x^{11})$	$(1, x^{15})$	6 / 6	2	every T	–
`c48c2`	$[[48, 4, 8]]$	4	standard	$(1, x^{11}, x)$	$(1, x^{15})$	7 / 7	1	every T	–
`t30a`	$[[30, 8, 5]]$	5	tree	$(1, x, x^4)$	$(1, x^6, x^{13})$	5 / 5	0	MC, $T \leq 6$	Z, X
`t30a2`	$[[30, 8, 5]]$	5	tree	$(1, x, x^4)$	$(1, x^6, x^{13})$	5 / 5	0	MC, $T \leq 6$	–
`t30b`	$[[30, 8, 5]]$	5	tree	$(1, x, x^4)$	$(1, x^6, x^{13})$	5 / 5	0	MC, $T \leq 6$	–
`t30b2`	$[[30, 8, 5]]$	5	tree	$(1, x, x^4)$	$(1, x^6, x^{13})$	5 / 5	0	MC, $T \leq 6$	–

Evidence classes. We label each statement. *Proven*: a hand proof, given in full in the supplementary material and audited once by an AI agent that did not write it (Sec. IX). *Machine-checked*: a certificate accepted by the checker, or a finite hypothesis checked by a script whose negative controls must and do fail. *Kernel-checked*: a theorem accepted by the Lean kernel on the standard axioms. *Measured*: an uncertified computation. A result *proven from machine-checked premises* is a hand proof whose hypotheses and finite inputs are machine-checked; every every-T claim in Table I has this class. How each check was run, and its controls, is in the verification log (App. A).

II. CIRCUITS, FAULTS AND THE DISTANCE

A. Detector error models

A stabilizer circuit carries Pauli channels at declared locations. A *fault* is one non-identity component of one channel. Its *symptom* is the pair (detectors flipped, observables flipped), computed by propagating the Pauli through the circuit. A *mechanism* is a distinct nonzero symptom; Stim's detector error model (DEM) lists them [8]. Write D_m and L_m for the detector and observable vectors of mechanism m over GF(2).

Definition 1. *For $X \subseteq \mathcal{M}$ write $D(X) = \sum_{m\in X} D_m$ and $L(X) = \sum_{m\in X} L_m$. A set with $D(X) = 0$ and $L(X) \neq 0$ is* undetectable and logical*; we call it a* solution*. The circuit-level distance of a circuit with mechanism set $\mathcal{M}$ is*

$$d_{\text{circ}} = \min\{|X| : X \subseteq \mathcal{M},\ D(X) = 0,\ L(X) \neq 0\}.$$

Mechanisms and faults. The minimum over mechanisms equals the minimum over faults for every circuit here. A fault set produces at most as many mechanisms as it has members, because equal symptoms cancel in pairs. Conversely, a minimum-weight solution is realized by at most as many faults, at most one per location (one channel instance): take one component per mechanism and combine those taken at one location into their product, whose symptom is the sum of theirs. That product is not the identity, or those members could be dropped, and the same channel contains it, because every channel here is a single Pauli flip, with one component, or a depolarizing channel, whose components are all the non-identity Paulis on its qubits.

Rates do not matter. d_{circ} depends on which Paulis can occur where, not on their probabilities. Every statement below is about the fault set of the stated circuit and noise placement.

B. Memory experiments

A memory-b experiment, $b \in \{X, Z\}$, prepares the data in the b basis, runs T rounds of extraction and reads the data out in the b basis. Detectors compare consecutive outcomes of each check; the b-type checks also have a first-round detector and a final detector that closes against the data readout. The observables are all k logical operators of type b, and a set is logical if it flips any of them, so d_{circ} is a distance for the whole logical block. We add no redundancy meta-checks. For the two morphing circuits of Ref. [3] the detectors and observables span every deterministic parity of the circuit except exactly 6 meta-checks of the other type's first-round outcomes, one per redundant check of that type (rows of its check matrix minus its rank; measured against Stim's tableau), and their recorded witnesses survive those meta-checks; for the other circuits we did not test them. Extra detectors can only remove solutions, so every lower bound here holds with meta-checks added.

C. The circuits

IBM's depth-8 cycle. The schedule of Ref. [1], with one ancilla per check. The tests re-derive its term maps from the matrix construction in IBM's public code (`decoder_setup.py` in `https://github.com/sbravyi/BivariateBicycleCodes`).

The three-colour cycle. The cycle of Fig. 12 of Ref. [2] in its version 2, whose figure states every CNOT, preparation and measurement time (version 1's figure has other labels and no preparation or measurement times); our schedule is that timetable. Its geometry is derived algebraically and matched to the long-edge directions of their Fig. 10; Stim accepts every detector in both bases; and the hook residuals reproduce all 6 patterns of their Fig. 11. The cycle is pipelined: a round's X checks finish during the next round's Z checks.

Morphing circuits. A morphing circuit has no ancillas. The $2n$ qubits of a mid-cycle BB code alternate between two contraction circuits, and each round ends in an n-qubit end-cycle code. We use the [[72,12,6]] circuit (mid-cycle [[144,12,12]]) and the [[144,12,12]] circuit (mid-cycle [[288,12,18]]) of Ref. [3]. The schedule is ported from the authors' builder (`github.com/Mac-Shaw/morphing_qec_circuits` at commit `1ee705f4`) and compared with it gate by gate: gates, measurement lists and every stabilizer's contraction round are identical for both codes and all 3 homomorphisms, and a schedule with two layers swapped is caught. The detectors are derived by stabilizer flow and declared wherever a value is determined.

Noise placement. IBM's cycle: noiseless data initialization; a flip after each ancilla reset; single-qubit depolarizing noise after each Hadamard; after each CNOT layer, two-qubit depolarizing noise on its pairs and single-qubit depolarizing noise on every idle qubit; a flip before each measurement and before the data readout. Three-colour cycle: a flip after data preparation and after each ancilla preparation; two-qubit depolarizing noise after each CNOT; a flip before each measurement; single-qubit depolarizing noise on idle data qubits in each time step (idle ancillas are not faulted). Morphing circuits: a flip after data preparation; after each reset of the right qubits a flip, and single-qubit depolarizing noise on the left qubits; after each CNOT layer, two-qubit depolarizing noise on its pairs (every layer covers every qubit, so there are no idle locations); flips before the measurements and single-qubit depolarizing noise on the left qubits after them; a flip before the readout.

Equivalence with SD6, and with Pan's circuits. These placements are not written as the standard depolarizing model SD6 of Gidney *et al.* [21], which faults every idle qubit in every time step and every preparation.

We rebuilt each circuit with SD6 noise instead and compared the sets of distinct (detector, observable) symptoms: for IBM's cycle, the three-colour cycle and both morphing circuits of Ref. [3], in both bases and at every $T = 1, \ldots, 4$, the full and the projected mechanism sets are identical (32 comparisons, each with a control that must and does change the set). We then built Pan's own circuits from his public release, with his SD6 implementation: his circuit files for IBM's cycle and for the three-colour cycle (the circuit of his Corollary 1), and his morphing builder [15]. In all 24 comparisons (four circuits, both bases, $T = 1, \ldots, 3$) his projected mechanism set equals ours after an invertible change of observable basis, so the same sets of mechanisms are undetectable and logical; each of the 8 near-miss controls (a legal CNOT-layer swap, one of his retimings of the three-colour cycle, his other layer order of the morphing circuits) must and does differ (App. A). The distances here are therefore those of Pan's circuits under SD6 at these T; we expect equality for every T, because the repeated block is the same, but we have not proved it, and we did not run either comparison for the circuits of Table II.

III. CERTIFIED LOWER BOUNDS

A. Same-type projection

Lemma 1 (projection)**.** *In a memory-b experiment, keep only the b-type detectors and let d_{proj} be the distance of the projected model. Then $d_{\mathrm{proj}} \le d_{\mathrm{circ}}$.*

Proof. Project the members of a full solution, drop those that become zero, and cancel equal pairs. The observable parts are untouched, so the result is a projected solution with no more members. □

The projection is the representation change that makes the search feasible. For IBM's cycle at $T = 2$ in memory-Z it takes 8,568 mechanisms to 1,584 over 216 detectors (8,640 to 1,584 in memory-X, and 7,800 to 1,584 for the three-colour cycle). Each projected detector is touched by 16 to 35 mechanisms, against about 148 in the full model, and each projected mechanism touches at most 6 detectors. The projection is also exact when every projected mechanism has a pure-type fault component, which flips no other-type detector, so that a projected solution lifts at the same weight. For the morphing circuits, which apply only CNOTs, this is proved as part of Theorem 2. IBM's cycle, and the three-colour cycle in memory-X, also apply Hadamards, which exchange a fault's X and Z parts, so that proof does not cover them; for IBM's cycle and the three-colour cycle the property is measured only, on two-round models (IBM's schedule on the [[72,12,6]] code in memory-X, the three-colour cycle in memory-Z). No claim below needs exactness, because every upper bound is a witness verified on the full model.

B. Anchoring on translation orbits

A BB code lives on $\mathbb{Z}_l \times \mathbb{Z}_m$ (for the gross code $l = 12$, $m = 6$). Translations act on qubits, and each circuit here is invariant under a group of translations, which therefore permutes its mechanisms and detectors. Observables are not invariant: a translated logical operator is another representative, so a mechanism's observable vector changes by stabilizer terms, and per-mechanism linearity fails.

Lemma 2 (affine observables)**.** *Let G be a group of translations with matrices M_g, K_g such that $L_{gm} = M_g L_m + K_g D_m$ for every mechanism m and every $g \in G$. Then a set X is a solution if and only if gX is.*

The map is linear in the pair (L_m, D_m); we call the observables *affine* because, for a fixed syndrome, L_{gm} is an affine function of L_m.

Proof. For an undetectable X, $\sum_{m \in X} L_{gm} = M_g \sum L_m$, and g permutes detectors, so gX is undetectable. Applying the same identity with $g^{-1} \in G$ to gX returns X, so $\sum L_m \neq 0$ if and only if $\sum L_{gm} \neq 0$. □

Number the orbits of G on mechanisms. Every solution has a translate that contains the representative of its smallest orbit o and uses only orbits $\ge o$. The search therefore starts once per orbit, from its representative, and allows only mechanisms of orbits $\ge o$. The groups are all 72 translations for IBM's cycle, the 24 colour-preserving ones for the three-colour cycle, and for the morphing circuits the kernel of the homomorphism (36 and 72 translations), because round $r+1$ is round r translated. Freeness of the action is not needed for soundness; the checker tests it anyway.

C. The search

A search state is a residual syndrome s, a logical flip ℓ, a remaining budget r and an orbit floor o.

Branching: If $s \neq 0$, pick any firing detector d. A completion contains an odd, hence positive, number of mechanisms touching d, so branching over the allowed mechanisms touching d loses nothing; a firing detector with none makes the state infeasible. At $s = 0$ the branch ends: it is a solution if $\ell \neq 0$.

Packing bound: If no allowed mechanism touches two detectors of a set U of firing detectors, a completion needs at least $|U|$ mechanisms.

Cover bound: Let c_d be the largest overlap $|\mathrm{supp}\, D_m \cap s|$ over allowed m touching d. With $Q = 2520$, the integers $y_d = \lfloor Q/c_d \rfloor$ satisfy $\sum_{d \in s \cap m} y_d \le Q$ for every allowed m, so every completion has at least $\lceil \sum_d y_d / Q \rceil$ members. A later version raises each y_d by the least slack of an allowed mechanism, an exact-integer dual ascent that keeps feasibility.

Leaves: Completions with at most two members are decided exactly by lookup, and in the latest format completions with at most three members are decided without a trace bit.

The search engine also keeps a transposition table keyed on the full state, which is sound because successors depend only on the state; the proof-producing prover omits it.

D. Certificates and the checker

The prover writes one bit for each node with nonzero syndrome and remaining budget at least three (four in the latest format): expanded, or pruned by a bound. Three formats appear below: the original IBM campaign's, 4 bytes per such node; one bit per node (CDP3, the format of the earlier prover of App. A); and CDP5, one bit per node of budget at least four, the three-member completions being decided by the checker without a bit. The checker, a separate C program, re-derives everything else from the model: the symmetry (invariance of the mechanism set, the affine observable map by Gaussian elimination over GF(2), the orbits and their representatives), the branching detector by a fixed rule (any firing detector is sound; a prover that branched elsewhere desynchronizes and is rejected), the child lists, every bound in exact integers, and every leaf with its own lookup. Tests make each of its rejections fire (App. A). The checker does not read a model hash; each run record binds a certificate to the SHA-256 of the exported model it was checked against, and the claims refuse a record whose model is not the one its search ran on. In the latest format, CDP5, the memory-Z proof for IBM's cycle at $T = 2$ and weight ≤ 9 is 92 KB, 18.3 times smaller than the 1,691 KB of the one-bit-per-node format (21.3 times for a three-colour model at weight ≤ 8), and checks in seconds (Table III).

E. Upper bounds

Witnesses come from the same engine run at weight d_{circ}. Each is lifted from the projected to the full DEM by choosing one fault component per mechanism, and verified there: it fires no detector, flips a logical, and every proper subset obtained by deleting one member fails.

F. What is trusted

Stim parses the instruction lists. Its error analysis is not trusted: an independent backward Pauli-sensitivity pass derives every model's fault set a second time, and the two sets are equal on all models the claims rest on: the 66 models of the IBM, three-colour and Ref. [3] circuits (432 to 66,912 mechanisms) and the 210 models of the circuits of Table II (180 to 12,624), every round count their searches, certificates and window checks used; a copy of the circuit with one noise line dropped, or one CNOT reversed, breaks the equality. The trusted base is the instruction lists, the projection (a few lines), the exporter's assignment of detectors to rounds and its time ordering, on which the anchoring by time shifts depends (cross-checked where the window checks bind the exported models), the arguments of this section, and the checker, whose calculus is proven sound in Sec. V.

Every certificate the results rest on was produced and checked again by the committed prover and checker, the version the Lean model follows: the 22 of IBM's cycle and 282 others, all verified on models byte-identical to the ones their searches ran on (Table IV). The small proofs ship in the repository; the 30 large ones are assets of the same release (Sec. IX) and also regenerate from their shipped models, and each of those has been produced twice, identically (App. A).

IV. EVERY NUMBER OF ROUNDS

A certificate covers one model. A claim for every T needs an argument that a long experiment has no lighter solution than a short one.

A. Time windows

In the IBM and three-colour models every mechanism touches detectors in at most two consecutive rounds. A minimum-weight solution therefore occupies a contiguous block of rounds: if no member started in some round strictly between the first and the last, the members before and after it would form two undetectable sets, one of them logical and lighter. The members of a minimum-weight solution of weight $\leq w$ start in at most w consecutive rounds. If interior rounds are time translates of each other and the boundary rounds do not depend on T, a solution in a long model translates into a short one. Two rules use this.

Rule W. No solution of weight $\leq w$ for $T = 1, \ldots, w+2$ implies none for any T.

Rule S (time shift). With orbits numbered in round order, let X be a minimum-weight solution of weight $\leq w$ whose first round is $r_0 \geq 2$. It is a time translate of a solution that starts in round 1 of the model with $T - r_0 + 1$ rounds; if that model has more than $w + 2$ rounds, the translate touches only rounds $\leq w+1$ and embeds, shape for shape, into the model with $w + 2$ rounds. So it suffices to certify, for $T = 1, \ldots, w + 2$, that no *irreducible* solution of weight $\leq w$ (no nonempty proper subset undetectable) starts in round 0 or 1: X is irreducible because it has minimum weight, and so is each translate. Each such certificate anchors only in two rounds.

TABLE III. Certificate sizes and times for representative models. *Mechanisms* are those of the projected model; w is the certified weight bound. The first two rows are certificates of the original IBM campaign, on which nothing in the results rests (App. A); the CDP5 rows are later benchmark runs on the same two models. The three-colour and morphing rows are the CDP5 proofs of the re-certification, written and checked by the committed prover and checker (Sec. III F). Times are wall-clock on one workstation with several threads on a loaded machine, not controlled benchmarks.

model	mechanisms	w	proof	prove (s)	check (s)
IBM, $T = 2$, Z	1,584	9	54.0 MB	14.5	78.9
IBM, $T = 2$, X	1,584	9	100.5 MB	19.9	169.7
IBM, $T = 2$, Z, CDP5	1,584	9	92 KB	2.8	2.7
IBM, $T = 2$, X, CDP5	1,584	9	168 KB	4.6	4.5
three-colour, $T = 2$, Z	1,584	10	21 MB	1,390	1,337
three-colour, $T = 2$, X	1,584	10	24 MB	1,503	1,392
three-colour, shift, Z (×12)	864–8,784	10	5.2–15.2 MB	6.4 h	4.7 h
three-colour, shift, X (×12)	864–8,784	10	6.6–34.0 MB	12.1 h	9.0 h
morphing $[[144, 12, 12]]$, $T = 1$, Z	864	11	24 MB	121	97
morphing $[[144, 12, 12]]$, $T = 1$, X	864	11	21 MB	114	93
morphing $[[144, 12, 12]]$, $T = 2$, Z	1,584	11	53 MB	386	311
morphing $[[144, 12, 12]]$, $T = 2$, X	1,584	11	50 MB	374	296

TABLE IV. The re-certification, per proof set. *Shipped*: in the repository; the other proofs of a set are release assets (Sec. IX). *Regen.*: regenerated by the committed prover; = *record*: byte-identical to the originally recorded proof; *old prover*: recorded proofs that the prover of development commit `72d03874` reproduces bit for bit (App. A); *2nd run*: regenerated again at another thread count, identically. Three-colour time-shift proofs equal to their records are (a), the others (b). Every proof was verified by the committed checker on a model byte-identical to the one its search ran on. Every count is re-derived from the records by `make_numbers_distance.py`.

proof set	proofs	shipped	regen.	= record	old prover	2nd run
2026 circuits (Table II)	238	238	238	238	0	0
morphing $[[72, 12, 6]]$, shift	14	14	14	0	14	0
three-colour, shift (a)	14	0	14	14	0	2
three-colour, shift (b)	10	0	10	0	10	10
three-colour, $T = 2$	2	0	2	0	2	2
morphing $[[144, 12, 12]]$	4	0	4	0	4	4
total	282	252	282	252	30	18
IBM, rule S (own campaign)	22	22	22	–	–	–
IBM, rule W (original)	22	0	–	–	–	–

Both rules need three window preconditions on the consecutive pairs $(T, T + 1)$ they span, rule W on every pair from its starting T on and rule S on every pair except that the first may satisfy only a weaker time-shift form of P3 (Theorem 1): (P1) each mechanism spans at most two consecutive rounds; (P2) the mechanism shapes that start in interior rounds form one set; (P3) the shapes that start in the first round and in the last two rounds do not depend on T. These can be checked on finitely many models. For every T they need a theorem.

Theorem 1 (window formula)**.** *Write the circuit as $C_T = P \cdot F \cdot B^{T-1} \cdot E \cdot R$: preparation, a first block, $T-1$ copies of a body block, a tail and the readout (the three-colour cycle is pipelined, so F lacks the previous round's X-check tails and E finishes the last round's; for IBM's cycle $F = B$ and E is empty). All maps below are linear over* GF(2) *on Pauli frames. Let A be the map that carries a frame through B; μ the map from a frame entering B to the flips of B's measurements of memory-type ancillas; and N and N_{obs} the maps from a frame entering the last body block to the flips of the memory-type detectors that read records of E or R, and of the observables. Assume that the builder emits C_T for every T with the closed-form detectors and observables of the supplement (`WINDOW_PROOF.md`, Sec. 1.3); that P measures nothing, B every ancilla once, F some ancillas and E the others once each, and R every data qubit once; and that there are integers $h, e \geq 0$ with $\mu(A + I)A^h = 0$, $NA^e = 0$ and $N_{\mathrm{obs}}(A + I)A^e = 0$; take the smallest such h and e. Then, with $U = \max(h, e) + 1$, start index $j_s = 1$ and $T_0 = j_s + U$, there are finite sets S, I,* End *independent of T such that for every $T \geq T_0$ the projected model is*

$$\mathcal{M}_T = S \cup \bigcup_{i=j_s}^{T-1-U} \tau_i(I) \cup \tau_T(\mathrm{End}),$$

where τ_i shifts detector rounds by i. The full model has a formula of the same form when μ, N cover the ancillas and detectors of both types, with $j_s = 2$ if F leaves an other-type ancilla unmeasured, else $j_s = 1$. Beyond a

threshold $T_1 \geq T_0$, P1–P3 hold for every pair $(T, T+1)$ of projected models; this needs the formula and three finite checks on the model at T_1: every mechanism spans at most two rounds (P1), and the windows of the rounds next to the start, and of those next to the end, equal the interior window set. The pairs below T_1 are checked on the exact models. For memory-Z the pair $(1, 2)$ satisfies only the weaker time-shift form, which is all rule S needs there.

The proof composes frames block by block: a fault in body copy i sees the same suffix $B^{T-i} \cdot E \cdot R$ in every C_T, and after U blocks its detector response stops changing. The formula alone does not give P1 or the windows near the boundaries: those are read off S, I and End at T_1, which the supplement does for each circuit and basis (`WINDOW_PROOF.md`, Corollary 1). For IBM's cycle the identities hold with $h = e = 0$ for both models: every block resets all ancillas first, so data Paulis pass a round unchanged and the readout closes the last round exactly; $F = B$ measures every ancilla, so $T_0 = 2$ for the projected and the full model alike. For the pipelined three-colour cycle they hold with $h = e = 1$ and fail at 0; $T_0 = 3$ for the projected model and 4 for the full one, whose start index is 2 because F leaves some other-type ancillas unmeasured. Sec. VI uses the full-model formula (`WINDOW_PROOF.md`, Theorem 1) to place witnesses, which live in the full model.

Status. Theorem 1 is a hand proof. Its hypotheses, the finite checks at T_1 and the pairs below it are machine-checked for both circuits and both bases, each check with a negative control that must fail, and the closed form agrees with Stim's DEM well beyond every T searched; an independent AI-agent review refuted nothing (App. A). That the builder emits C_T for every T is a proof by reading the code (`WINDOW_PROOF.md`, App. A), confirmed mechanically at 12 values of T.

B. The fold theorem for morphing circuits

A two-round morphing circuit is a palindrome around each measurement: contract with F_r, measure the right qubits, reset them in the same bases, expand with F_r^{-1}. Every fault therefore has the symptom of a Pauli P at a mid-cycle point t, its *mid-cycle image*. Write $\sigma_S(P)$ for its syndrome against the memory-type generators, $\sigma_L(P)$ for its flips of the memory logicals, and ι_t for the map that places generator g's bit on the one detector of g whose time window contains t.

Theorem 2 (fold)**.** *For a two-round morphing memory experiment, assume: the builder emits it for every number of rounds with the closed-form memory-type detectors and observables of the supplement (`FOLD_PROOF.md`, Sec. 1.4) and with other-type detectors that read only outcomes measured in the other basis; each CNOT layer, of any number per contraction, uses each qubit at most once; every right qubit is measured and reset each round; carried checks and logicals touch only right qubits measured in their own basis (trackability); the mid-cycle images of the noise components after each expansion layer are, as a set, those of the noise components after its mirrored contraction layer, and the images of the reset noise are those of the measurement noise (palindrome); preparation and readout noise have their images among those of the reset and measurement noise; and the part of each fault that flips memory-type measurements (its X part in memory-Z) is itself a fault. Then a fault whose mid-cycle image is P at t has projected symptom $(\iota_t(\sigma_S(P)), \sigma_L(P))$, the projected model is*

$$\mathrm{Proj}_T = \bigcup_{t=0}^{T-1} \iota_t(\mathrm{Proj}_1),$$

and for every $T \geq 1$, $d_{\mathrm{circ}}(T) = d_{\mathrm{proj}}(T) = d_{\mathrm{proj}}(1) = d_{\mathrm{circ}}(1)$.

The palindrome hypothesis is about images, not raw channels: the noise after a contraction layer is pulled back through that layer and the noise after the mirrored expansion layer is not, so the two image sets agree when each layer's noise set is invariant under conjugation by that layer, as for the depolarizing noise here. Equal raw noise on mirrored layers is not enough. Independent X flips on the support of one logical, after the same layer in the contraction and in the expansion of the [[72,12,6]] circuit, give a one-round model with no solution and a two-round model with one; the image check refuses that noise (supplement, `FOLD_PROOF.md`, and a test).

The map that sums each generator's detectors undoes every ι_t at once and is linear, so a solution of the T-round model folds onto one of the one-round model with no more members; ι_0 is injective, so a one-round solution embeds back; and that part of each fault lifts projected solutions to full ones at the same weight. The hypotheses are statements about two rounds and the boundaries, so none depends on T. Shaw and Terhal prove $d_{\mathrm{circ}}(T) = d_{\mathrm{circ}}(1)$ for two-round morphing circuits by time-reversal folding in their convention (ideal preparation and readout, with meta-check detectors; [4], Prop. 1). Pan's record E-017 checked a basis-free form of their fold premise by computation for the [[144,12,12]] circuit with noisy preparation and readout, at two to five rounds, and argued every T by periodicity [15], Sec. 3. Theorem 2 states that case as a theorem for any two-round morphing circuit with one explicit linear map, and also holds with meta-checks added.

Status. Theorem 2 is a hand proof, written for any number of involutive CNOT layers per contraction; every circuit it is applied to has 3 layers per contraction, the number the checking script requires. A script checks all six hypotheses, the palindrome as images, for both circuits, both bases and all three homomorphisms, and for each circuit of Table II with a standard schedule, with negative controls that must and do fail; it identifies the builder's circuit with the template, at 10 values

of T for the two circuits and at every $T \leq 32$ for those of Table II, where every component's projected symptom equals the formula above; and an independent AI-agent review checked the union formula against Stim's DEM alone at fresh points and on small morphing codes the proof never saw, refuting nothing (App. A). That each builder emits the template for every T is a proof by reading its code: the morphing builder of Sec. II C and the generalized builder of Sec. VI D (`FOLD_PROOF.md`, Apps. A and C).

V. LEAN

A. The calculus

The certificate calculus is formalized in Lean 4 (v4.34.0) with Mathlib (v4.34.0). A DEM is a pair of maps from mechanisms to vectors over $\mathbb{Z}/2$; a solution is a finite set of mechanisms whose syndromes sum to zero and whose observables do not. A valid tree has the constructors of Sec. III: done, spent, leaf, prune (under each bound) and expand at any firing detector; the three-member terminal is proved to be a valid tree. The main theorem reads:

Theorem 3 (`cert_sound`)**.** *If the incidence tables contain the true incidence, the translations form a group that preserves the mechanism set and maps observables affinely, every mechanism's orbit label is below the number of anchors, and for every orbit the tree rooted at its representative is valid with budget $w - 1$, then every solution has more than w members.*

A scoped form, `cert_sound_scoped`, gives rule S on a single model: no irreducible solution of weight $\leq w$ touches a round $\leq$ the declared bound. The theorems also cover the packing and cover bounds in the checker's exact integers, the dual ascent (every raise keeps the dual feasible), the terminal and the leaves, and anchoring with affine observables. Concrete models run the chain end to end: one with a non-trivial group and observables that map only affinely, and another on which only the dual ascent prunes, where the C checker agrees. The receipt lists 40 theorems, each on the standard axioms (`propext`, `Classical.choice`, `Quot.sound`), pins every source by hash, and a Lean-free gate refuses a stale source, a `sorry` or a non-standard axiom.

B. What the formalization found

Formalizing the rules exposed defects in the C programs and an overstated verdict. A model in which one mechanism lists a detector twice XORs that detector away while its degree still counts it. On such inputs, with crafted proofs, the checker printed VERIFIED for false claims. Every program now refuses a repeated detector at load. Separately, the time-shift verdict claimed "no solution" where the scoped proof establishes "no irreducible solution"; the verdict now says so, and a Lean example shows on a concrete model that the stronger reading is false. None of this touched a recorded result: no certified model can contain a repeated detector, because the independent derivation of Sec. III F writes each symptom as a set and its mechanism list equals the exporter's on every model the claims rest on; and rule S needs only the irreducible form (details in App. A).

C. Distances checked by the kernel

The step from "the C checker printed VERIFIED" to "the Lean hypotheses hold" is argued line by line, not proved. For small models we remove it. A Lean program performs an exhaustive anchored search over the exported projected model, through an incidence table that is supplied as data and checked for completeness, and a theorem (`Exec.cert_sound_exec` and its table form) states that the program's acceptance implies the distance bound, through the same `cert_sound`. A generated module per model runs the program in the kernel, one `decide +kernel` declaration per anchor, combined by rewriting so that no single declaration grows, and adds a witness at the next weight. The result is a theorem that the exported model's distance is exactly d, on the standard axioms, with no C code and no compiler trusted in the check; the model itself still comes from Stim and the exporter, whose output the independent derivation of Sec. III F confirms. The 4 theorems are the one-round projected models of `c42a` (210 mechanisms, $d_{\text{circ}} = 6$) and `t30a` (180 mechanisms, $d_{\text{circ}} = 5$) in both bases (Table II); none is a gross-code model. Each module's model literal is regenerated from a fresh export whose hash must equal the certified one. The kernel needs 141–624 s per model at up to 2.6 GB. For `c42a`, Theorem 2 then carries the kernel-checked one-round distance to every T; for `t30a`, a tree circuit, it covers $T = 1$ only. Gross-code models, with 1,584 projected mechanisms at $T = 2$ and weights of 9 or more, would need the bounds and the symmetry inside the kernel program; that is future work.

D. What is not proved

The correspondence between the C checker and the Lean predicate, including the trace decoding, the leaf index, the elimination, parsing, arithmetic and the compiler, is argued, not proved; an executable cross-check holds the checker's own bound, terminal and leaf procedures against breadth-first brute force on random models, and one-token mutants of those rules are caught (App. A). Rule S across models, and everything upstream of the DEM (the circuits, the DEM derivation, the projection, Theorems 1 and 2, witness lifting), is outside the formalization.

VI. RESULTS

A. IBM's depth-8 cycle

22 time-shift certificates, written by the committed prover in CDP5 and verified by the committed checker (the version the Lean model follows), state that no irreducible undetectable logical set of weight ≤ 9 starts in round 0 or 1, for $T = 1, \ldots, 11$ in both bases. With rule S and Theorem 1 (for memory-Z the pair $(1, 2)$ satisfies the weaker time-shift form, which is all rule S needs) they give $d_{\text{circ}} \geq 10$ for every T. The proofs are 24.0 to 232.8 KB, check in 1.5 to 24.6 s, and all ship with their models; each shipped model is byte-identical to a fresh export of the cycle.

A weight-10 witness of end-boundary type in each basis, verified on the full DEM at $T = 2$, lies in $\tau_T(\text{End})$ of the full formula, whose threshold is $T_0 = 2$, and so occurs in every model with $T \geq 2$; it was also checked directly at $T = 2, \ldots, 6$. Hence $d_{\text{circ}} = 10$ for every $T \geq 2$, memory-Z and memory-X, 2 below the code distance 12. In our fault placement the bound $d_{\text{circ}} \leq 10$ of Ref. [1] is attained. At $T = 1$ the certificates give only $d_{\text{circ}} \geq 10$. The value agrees with Crocker's exhaustive search for twelve noisy rounds [10] and with Pan's records: VeriPB certificates at every $R = 1, \ldots, 12$ under SD6 and IBM's own fault model [12], and every number of rounds, under SD6 from his bulk criterion [13] and for his re-implementation of IBM's fault model from the certified range and a written reduction [14], on a circuit whose projected mechanism sets equal ours at $T = 1, \ldots, 3$ (Sec. II C); and with Cheng, Guo and Pan [17], who certify 10 with their exact solver, in both bases for one noisy cycle of the memory instruction of IBM's architecture study and for twelve noisy cycles of the circuits of IBM's public code, and as the smaller of the two bases for six-cycle memory experiments in their own noise model (their Supplementary Secs. 8.1 and 9.2). The bound of Ref. [17] that holds for every number of rounds, its time collapse, is 8 in memory-X and 9 in memory-Z; ours rests at every T on the certificates above and Theorem 1, whose finite hypotheses are checked by script.

B. The three-colour cycle

At $T = 2$ the search found no solution of weight ≤ 10 in either basis, certified, and weight-11 witnesses in both bases were verified on the full DEM. For every T we used rule S: 24 time-shift certificates, $T = 1, \ldots, 12$ in both bases, each on exactly the model the time-shift search ran on (864 to 8,784 mechanisms, 36 to 60 anchors, proofs of 5.2 to 34.0 MB), all verified by the committed checker. Theorem 1 supplies the window preconditions for every pair. The witnesses are of start type: every member lies in the start part S of the full formula, so they occur in every model with $T \geq 4$, the full model's T_0, and they were checked directly for $T = 2, \ldots, 9$ and on the full DEM for $T = 2, \ldots, 12$. Hence $d_{\text{circ}} = 11$ for every $T \geq 2$ in both bases: the conjectured value of Ref. [2] holds for the circuit of their Fig. 12 (v2), under the noise placement of Sec. II C, as Pan also found under SD6, from the reduction of Ref. [2] and exhaustive searches [11], on a circuit whose projected mechanism sets equal ours at $T = 1, \ldots, 3$. Cheng, Guo and Pan [17] also confirm the conjecture, 11 in both bases, with their exact solver at a fixed number of rounds, on a circuit rebuilt from the text of Ref. [2] (their Supplementary Table 8); their repository's records, made at six rounds, give the time-collapsed bound as 11 in both bases, and by their Supplementary Sec. 5.4 that bound holds for every number of rounds. We have not compared their circuit with ours. The cycle loses 1 against the code distance. The size of a time-shift certificate stops growing with T once T exceeds the window: the memory-X proofs have identical trace length for every T from 5 to 12.

C. Morphing circuits

For the [[72,12,6]] circuit, 14 time-shift certificates of weight ≤ 5, $T = 1, \ldots, 7$ in both bases, with the space-time window preconditions checked on every pair (round $r + 1$ is round r translated, and observables are required to map affinely between paired windows), give the bound for $T \leq 7$; Theorem 2 with the one-round certificates gives it for every T. All were re-checked by the committed checker, and their proofs ship. Weight-6 witnesses lift, survive the meta-checks and embed into every $T = 2, \ldots, 7$. So $d_{\text{circ}} = 6$ for every $T \geq 1$: in our fault placement the BP-OSD bound of Ref. [3] is attained, as Pan's record E-019 reports by exhaustive search [16].

For the [[144,12,12]] circuit, certificates of weight ≤ 11 were verified at $T = 1$ and $T = 2$ in both bases by the committed checker (Table III). Weight-12 witnesses, found in the one-round model in 545 and 1,368 s, were verified in every $T = 1, \ldots, 13$ and survive the meta-checks. With Theorem 2, $d_{\text{circ}} = 12$ for every $T \geq 1$, in both bases, the value of Pan's record E-017 [15]. A second route, the fold map checked separately at each $T = 1, \ldots, 13$ (52 rows over both circuits and bases), agrees. The circuit has no ancillas and keeps its end-cycle code's distance 12. That end-cycle code has the gross code's parameters [[144,12,12]] but is a different code: its stabilizer group contains elements of weight 9 (read off the circuit's deterministic final detectors), while the gross code's generators have weight 6 and commute, so every element of its stabilizer group has even weight, and qubit permutations and local Cliffords preserve weight (as Pan's E-017, Sec. 4.2, also shows [15]). IBM's cycle and the three-colour cycle, which act on the gross code, lose 2 and 1; comparing them with the morphing circuit compares two codes with the same parameters, not two schedules on one code. Table II's CNOT-order pairs are the fixed-code comparisons: the two circuits of each

pair have the same mid-cycle code and, read off their deterministic final detectors, the same end-cycle stabilizer group, in both types and after either round parity (tested). Distance is not logical error rate, and we make no claim about the latter.

D. The 2026 morphing circuits

Ref. [4] searches two-round morphing circuits over Abelian two-block group-algebra codes and reports end-cycle parameters only; it names neither the groups nor the polynomials. We re-found codes with its stated parameters by search. Among the codes we searched on $\mathbb{Z}_{42}$, 4 standard-schedule circuits have end-cycle code [[42,6,6]] of degree 4. No standard-schedule circuit we searched on $\mathbb{Z}_{30}$ reaches [[30,8,5]]; we obtained it from their homomorphism-based circuits (their Def. 10) fixed by a single contraction tree (their Sec. 2.4), which we call tree circuits. For the parameters [[48,4,8]], which their Fig. 25 plots for a morphing circuit (read off the plot; the text does not list them), we ran 6 standard-schedule circuits on $\mathbb{Z}_{48}$. Whether the paper's own circuits are among ours cannot be decided from the paper. Each circuit is built in our convention by a generalized builder, which reproduces the morphing builder of Sec. II C byte for byte on the circuits of Ref. [3]; the standard schedules equal the authors' builder gate by gate, with a swapped-layer control that fails. Since B has two terms, the middle CNOT layer of each standard contraction covers half the qubits, and the idle ones get single-qubit depolarizing noise. That this builder emits Theorem 2's circuit for every T is read from its code (`FOLD_PROOF.md`, App. C), not inferred from its agreement on those two codes, and is checked at every $T \leq 32$.

Table II gives the results. For each circuit and basis, the one-round search finds weight d with a minimal witness verified on the full DEM, the same weight appears at every T up to $d+1$, certificates of weight $\leq d-1$ are verified at one and two rounds, and time-shift certificates and window preconditions hold at every T up to $d+1$; every certificate was re-checked by the committed checker, and every proof ships. For the standard schedules Theorem 2's hypotheses hold (checked with their controls), so the value holds for every T; for the tree circuits the class is machine-checked to $T = 6$ only. As Pan's records show for the codes of Ref. [3] [16], the CNOT order is not a detail: circuit `c42a` keeps $d_{\mathrm{circ}} = 6$, while the other [[42,6,6]] circuits lose up to 2. The degree-5 tree circuits for [[30,8,5]] keep their distance. No standard-schedule circuit we ran for the parameters [[48,4,8]] reaches 8; the best, 2 of 6, reach 7 (cyclic groups only; deeper trees not searched).

VII. SCOPE AND LIMITATIONS

Noise placement. Every distance is a property of the fault set of Sec. II C, which equals SD6's on the circuits and rounds compared there. We did not compare it with the noise models under which the papers that introduced the circuits reported their bounds, so "attained" and "the value they conjectured" compare numbers, each in its own convention. A different placement can change d_{circ}: more fault locations can only lower it, and extra detectors can only raise it, so our lower bounds carry over to a convention whose fault set is contained in ours and whose detectors contain ours. For the morphing circuits the fold theorem also holds with meta-checks, which their convention uses; that ideal preparation and readout leave the model unchanged is predicted by the theorem's boundary hypothesis but was not run. On the smaller [[72,12,6]] instance of IBM's cycle ($T = 2$, memory-Z), deleting every single-qubit channel and flip leaves the mechanism set unchanged (tested): two-qubit noise after the CNOTs already produces every symptom.

Transcription. The three-colour cycle is transcribed from the timetable of Fig. 12 in version 2 of Ref. [2]; Pan's circuit file for it gives the same projected mechanism sets (Sec. II C). That is agreement between two transcriptions of one figure and its authors' text, not a comparison with the authors' own circuit file, which we did not find. The 2026 circuits are re-found, not copied (Sec. VI D).

Hand proofs. Theorems 1 and 2 are hand proofs, given in full in the supplementary files `WINDOW_PROOF.md` and `FOLD_PROOF.md`, each audited once by an AI agent that did not write it; no human expert has read them. Their hypotheses are machine-checked, and the claim that the three builders emit the template for every T rests on reading their code (`WINDOW_PROOF.md`, App. A; `FOLD_PROOF.md`, Apps. A and C), confirmed at finitely many T. Neither theorem is formalized.

The checker. The calculus is proven sound; that the C program implements it is argued and cross-checked, not proved. Every certificate the results rest on was re-checked by the committed version (Sec. III F). The tree circuits of Table II are machine-checked only to $T = 6$; beyond that their every-T status rests on the window lemma's preconditions holding by periodicity, which we have not proved for them.

Distance is not performance. We report no logical error rates. Two circuits with the same d_{circ} can differ in the number of minimum-weight solutions and in their behaviour at realistic noise.

VIII. RELATED WORK

Circuit distances of these circuits. Crocker [10] and Pan [11–16] reported the gross-code-scale values concurrently with or before this work (Sec. I); Crocker's search uses the same three reductions as ours. Pan certifies the infeasibility claims for his new interleaved circuits with

SAT and pseudo-Boolean proofs, checked with drat-trim (with LRAT cores checked by cake_lpr) and VeriPB [22–24], a proof system independent of ours, and his records certify IBM's lower bound up to twelve rounds the same way, for a strengthened formula whose strengthening rests on a written argument (Sec. I). On all four circuits our projected mechanism sets equal those of his circuits at one to three rounds (Sec. II C), so the agreements are on one object. Cheng, Guo and Pan [17] write code and circuit distances as tropical partition functions and evaluate them with rank-based dynamic programs and a weight-bounded search that, like ours, branches on a firing detector, anchors on translation orbits and prunes with a covering bound. Their sector reduction is our same-type projection; their time collapse sums each check's detectors over the rounds and, for single-ancilla CSS circuits, is the extended-code bound of Ref. [2], a lower bound for every number of rounds; lifting its minimum solutions to circuit fault sets raises the bound for a fixed number of rounds. Their audit of 26 papers includes three that we cite [1, 2, 9]: it finds the upper bound 10 of Ref. [1] for the gross-code circuit exact and the conjecture of Ref. [2] true, and lowers to 16 the values 18 and 24 given for the two-gross circuit [1, 9], which we do not treat; no value stated here is among those it corrects. Specific to this paper, among the works we found, are certificates re-derived by a separately written checker under a calculus proven sound in Lean, the statements for every number of rounds from Theorems 1 and 2, certificates for the morphing circuits, which their paper does not treat, and the circuits of Table II. Wills *et al.* [18] believe a morphing idling circuit to have circuit distance 12. Exact circuit distances of smaller BB circuits were obtained with a MaxSAT solver on Stim's SAT export by Aghababaie Beni, Higgott and Shutty [7]. Webster, Jacob and Higgott benchmark exact and heuristic distance-finding algorithms for quantum codes and extraction circuits, including BB extraction circuits [9]. Jacoby, Retzker and Pastawski compute circuit-level distances of their stairway codes with an integer-programming decoder, reporting an upper bound where it does not finish [25]. Chen, Jafari and Lai compare SAT, MaxSAT and SMT formulations of the exact code distance, not the circuit distance, of QLDPC codes [26].

Certificates and formal verification. Branch-and-bound with linear-programming lower bounds and independently checkable branch-and-bound certificates are established in exact integer programming [27, 28], and certifying algorithms and program checking are classical [29, 30]; the method here specializes them to the fault hypergraph of a detector error model, with translation anchoring under affine observables and time-window reductions particular to periodic extraction circuits. Its certificates are about one bit per search node, and their checker's calculus, not only its output, is proven sound. An and Yang [31] certify code parameters by certificates a proof assistant's kernel decides by computation; for the gross code their kernel computation supplies the upper-bound witness and the dimension, and the lower bound on its distance is transported, as they state, from a separate Lean formalization (Jain's QECLean). That is the code distance, not the circuit distance. Formal verification of quantum error correction in a proof assistant is also the subject of Ehatamm *et al.* [32] and of Huang *et al.* [33], who verify QEC programs in a program logic formalized in Coq; our formalization covers a certificate checker for circuit distance, not a code, a decoder or a program. The circuits are those of Refs. [1–4]. Our literature search for every value called new here is logged in the repository (`docs/PAPER3_LITERATURE_SEARCH.md`); we make no priority claim beyond the values stated as new, and no claim about the order of discovery beyond the public dates.

IX. REPRODUCIBILITY

The code, the evidence and the Lean development are released under the Apache License 2.0 at `https://github.com/coherence-energy-labs/oneq-certificates` (release v2026.11.07.2); paths below are relative to that repository. The code is in `experiments/circuit_distance_exact/`. The claims file `evidence/circuit_distance_exact/results.json` is assembled from the run records beside it by `collect_results.py`, which refuses a claim whose records are missing or disagree; the test suite checks that the committed file equals a fresh assembly. The Lean development is `lean/CertR198`, with a receipt gate that needs no Lean. The 22 CDP5 proofs for IBM's cycle ship with their models in `evidence/circuit_distance_exact/ibm_recert_20261026/`, and the 252 re-certified proofs of the [[72,12,6]] morphing circuit and of Table II with theirs in `evidence/circuit_distance_exact/recert_20261102/`; each re-checks in seconds with `ccheck`. The 30 proofs of the three-colour cycle and the [[144,12,12]] morphing circuit, 0.73 GB together, are too large for the repository and are assets of the same release: `oneq-recert-proofs-3c.tar.gz` (26 proofs) and `oneq-recert-proofs-morph144.tar.gz` (4), 371 MB compressed, with every member's size and hash in `evidence/circuit_distance_exact/recert_20261102/release_assets.json`. With both downloaded into a directory `DIR`, `experiments/circuit_distance_exact/recert_assets.py verify --archives DIR --extract` checks each archive and each proof against that record and the manifest and puts the proofs where the manifest's cells are, and with `check` runs the committed checker on them, requiring each recorded verdict. A matching hash shows that a file is the proof the manifest names; only the checker decides it, and for these proofs that is slow: in the re-certification, at 8 threads, the checker took 1.5 min to 1.2 h per proof and 14.7 h for all 30, against at most 1.2 s for a shipped one. We ran this

route on the published archives: both verified, and the checker re-verified 5 of the proofs with their recorded verdicts (`evidence/circuit_distance_exact/recert_20261102/asset_checks/`); the other 25 were verified in the re-certification itself. The committed prover also regenerates each of the 30 from its model, bit for bit. Every exported model and claims file is written in one declared byte format on every platform (`serialization.py`), so a fresh export on any operating system reproduces the archived bytes its records bind. The cross-check against Pan's circuits is `pan_crosscheck.py`, which pins his files by hash. The supplementary files `WINDOW_PROOF.md` and `FOLD_PROOF.md` are the hand proofs of Theorems 1 and 2, as they stand in the same directory as the code. Models were exported with Stim 1.16.0; the C programs were compiled with GCC 16.1.0. Every number in this paper is a macro written by `paper/tools/make_numbers_distance.py` from those committed files; it refuses to run when the claims file differs from a fresh assembly of its records, when the Lean receipt does not certify the current sources, or when a certificate is bound to a model other than the one its search ran on. Times, sizes and ratios computed from the records are rounded to the nearest value at the digit shown, except that a maximum ("up to", "at most") is rounded up and a printed range is widened outward to contain the data; counts, weights and distances are exact. The public repository's history is squashed; the one development commit this paper names (App. A) is in a repository whose history is not public, and the file it names ships as the bytes that ran, with its sha256 in the run records.

Use of AI tools. AI tools assisted with software development, the proofs and manuscript preparation. The independent reviews in Sec. IV and the reviews of this manuscript archived with it (`docs/reviews/`) were carried out by AI agents that had not written the artifact under review. Of the four external audits of its releases of November 2026, also archived there, the third calls itself AI-assisted, the fourth states that it was done by an AI model with AI sub-agents, and the first two do not say who or what produced them. None of these is third-party peer review. The author is responsible for all content.

Appendix A: Verification log

This appendix records how each check of the main text was run; the main text states each claim once with its evidence class.

Checker rejections. Tests make every rejection of Sec. III fire: a raised weight claim, a flipped bit, truncation, trailing bytes, a fake prune, a prover that ignores solutions, a model that is not invariant, wrong orbit labels, and a model that lists a detector twice in one mechanism. The checker's own bound, terminal and leaf procedures are held against breadth-first brute force: on 4,798 queries over 20 random models no prune cut a completion and no terminal missed one, and each of the 9 one-token mutants of those rules produces violations (44 to 1,941).

Re-certification. The certificates were first checked by the checker version current when each was produced, several of them before the fixes of Sec. V B. Every certificate the results rest on was therefore produced and checked again by the committed prover and checker (Table IV). For those 282 every re-exported model is byte-identical to the model its original search ran on. The prover's output does not depend on scheduling: each search task writes its own buffer, and the buffers are written in task order. 252 of the regenerated proofs are byte-identical to the recorded ones: the 238 of Table II and 14 three-colour time-shift proofs, 54 of them carrying trace bits (up to 34 MB). The other 30 (14 of the [[72,12,6]] morphing circuit, 10 three-colour time-shift, 2 three-colour $T = 2$ and 4 [[144,12,12]] morphing proofs) were written by an earlier prover, before the proof language became CDP5, which omits trace bits for subtrees the checker decides itself; compiling the prover of an earlier commit (`72d03874`) and running it on the same model reproduces every one of those recorded proofs bit for bit, in the earlier format (CDP3). That commit is in the development repository, whose history is not public but is available to referees on request; its prover ships as `experiments/circuit_distance_exact/old_provers/cdist_prove_72d03874.c`, the bytes that ran, with the sha256 every reproduction record pins, and as that commit's source snapshot (`evidence/revision_snapshots.json`). The 252 proofs that ship in the repository are another set of that size: the 238 of Table II and the 14 of the [[72,12,6]] circuit. The 30 that are release assets instead (Sec. IX) are the 24 three-colour time-shift, 2 three-colour $T = 2$ and 4 [[144,12,12]] morphing proofs, a set that shares 16 proofs with the 30 written by the earlier prover; of them 14 equal their records and the other 16 were regenerated a second time at 4 and 6 threads instead of 8 (18 such runs, with trace bits, up to 54 MB), identically; so each has been produced twice. 212 of the 256 morphing-circuit proofs carry no trace bits at all: there the checker's verified terminal procedure repeats the whole search, and the proof adds little beyond its header. The three-colour re-certification took 32 h of proving and checking; in the original campaign the longest cell (memory-X, $T = 8$) took 2.8 h. The campaign's original 22 certificates for IBM's cycle used an earlier trace format, 4 bytes per node, and stated the stronger "no undetectable logical error of weight ≤ 9" through rule W, with proofs of 10.8 to 820.0 MB checked in 7.3 to 728.4 s; the checker version that verified them is not preserved in either repository, and nothing in the results rests on them. Rule S needs the mechanisms listed in round order, so only 2 of the 22 IBM re-certification models are byte-identical to the campaign's; the rest list the same mechanisms in another order, with the orbits numbered accordingly.

The window theorem's checks. A script checks the hypotheses of Theorem 1 for both circuits and both bases, each against a negative control that must fail (60 controls, all fail), confirms the builder's instruction stream against the template at 12 values of T up to 32, and compares the closed form with Stim's DEM alone at $T = 16$, 24 and 32, beyond every T searched: equal in every comparison. An independent AI-agent review rebuilt S, I and End from Stim at another reference round and compared the formula with Stim's DEM at T up to 27: 102 comparisons, no difference, every control failing.

The fold theorem's checks. A script checks all six hypotheses of Theorem 2, the palindrome on mid-cycle images together with the layer invariance that implies it, for both circuits and both bases with 20 negative controls per run (4 runs, all controls fail), and for the other homomorphisms with 18 controls per run. It identifies the builder's circuit with the template at 10 values of T up to 32, where every component's projected symptom equals the formula (5.7×10^6 components, no mismatch). Run on the circuits of Table II with standard schedules, built by the generalized builder, it finds every hypothesis in both bases, every one of its 18 controls per run failing, and identifies each circuit with the template at every $T = 1, \ldots, 32$ (4.3×10^7 components, no mismatch). An independent AI-agent review checked the union formula against Stim's DEM alone at 18 fresh points (circuit, basis, homomorphism, T), with both of its controls disagreeing at the 6 points where they were run, and ran 8 small BB morphing codes the proof never saw: their distances are constant in T, and breaking one hypothesis makes them drop. The weight-12 witnesses, sent through ι_0, are solutions at $T = 32$, with a one-mechanism-deleted control that fails.

The formalization's findings. On models in which a mechanism lists a detector twice, an anchor whose syndrome cancelled to zero received no task, and, in a faster leaf procedure under development, a filter that assumed degree equals syndrome weight misjudged a completion; the same input crashed a development version of the prover. None of the 825 model files then in the campaign directories had a repeated detector.

Pan's circuits. The cross-check builds, from Pan's release v2.0 (its data package, pinned by hash), his noiseless memory circuits for IBM's cycle and for the three-colour cycle from his circuit files, and his morphing circuits from his builder, adds his SD6 noise, and projects each detector error model onto the memory-type detectors. Detectors are matched by type, check support and round, never by index. For each circuit, basis and $T = 1, \ldots, 3$ it fits the map $L_{\text{ours}} = A\,L_{\text{Pan}} + KD$ over GF(2), requires A invertible, and requires the mapped set to equal ours: 24 of 24 are equal. Each of the 8 controls is a near miss that must differ and does: for IBM's cycle, two of its CNOT layers exchanged; for the three-colour cycle, one of Pan's own retimings that keeps the CNOT orders; for the morphing circuits, Pan's other layer order. Removing a single two-qubit noise line is not a usable control here: no such removal changes these projected sets, because other faults produce every symptom it removes.

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